

Program 8

Create Kubernetes Pods using YAML Descriptors

DevOps

- Program8
- nodejs-configmap.yml
- nodejs-pod.yml
- server.js

Step 1 : Create server.js file and add the following code

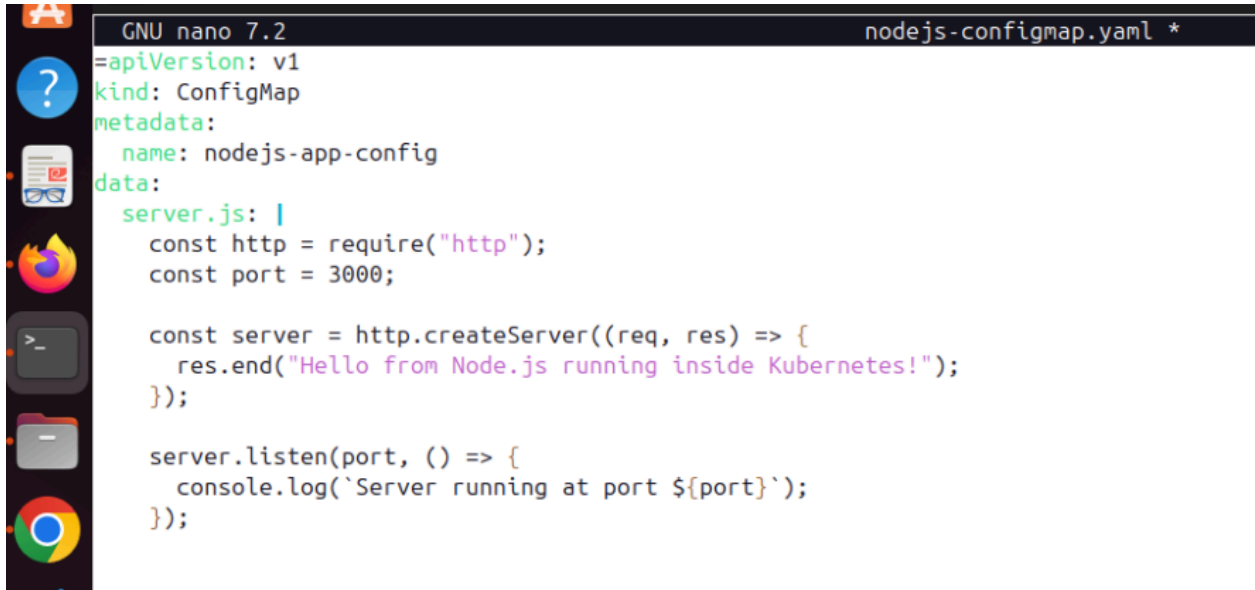
nano [server.js](#)

```
const http = require('http');
const port = 3000;

const server = http.createServer((req, res) => {
  res.end("Hello from Node.js running inside Kubernetes!!");
});

server.listen(port, () => {
  console.log(`Server running at port ${port}`);
});
```

Step 2 : Create nodejs-configmap.yml and add the following code

A terminal window with a dark background. On the left is a vertical sidebar with icons for a home folder, a question mark, a document with a red 'X', a pair of glasses, a Firefox browser, a terminal, a folder, and a Chrome browser. The terminal title bar shows 'GNU nano 7.2' and 'nodejs-configmap.yaml *'. The content is a YAML file for a ConfigMap.

```
GNU nano 7.2 nodejs-configmap.yaml *
=apiVersion: v1
kind: ConfigMap
metadata:
  name: nodejs-app-config
data:
  server.js: |
    const http = require("http");
    const port = 3000;

    const server = http.createServer((req, res) => {
      res.end("Hello from Node.js running inside Kubernetes!");
    });

    server.listen(port, () => {
      console.log(`Server running at port ${port}`);
    });
```

Step 3: Create Pod YAML

nano nodejs-pod.yml

A terminal window with a dark background. The title bar shows 'GNU nano 7.2' and 'nodejs-pod.yaml'. The content is a YAML file for a Pod.

```
GNU nano 7.2 nodejs-pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: program-8
  labels:
    app: nodejs
spec:
  containers:
    - name: nodejs
      image: node:18
      command: ["node", "/usr/src/app/server.js"]
      volumeMounts:
        - name: app-code
          mountPath: /usr/src/app
      ports:
        - containerPort: 3000
  volumes:
    - name: app-code
      configMap:
        name: nodejs-app-config
```

Step 4 : Apply the YAML files

```
kubectl apply -f nodejs-configmap.yaml  
kubectl apply -f nodejs-pod.yaml
```

Step 5 : Check pod status

```
kubectl get pods
```

Step 6 : Access the node.js app



```
1RV24MC106_Srusti_G_C@srusti-IdeaPad-Slim-3-15IAH8: ~/program8  
1RV24MC106_Srusti_G_C@srusti-IdeaPad-Slim-3-15IAH8$ kubectl port-forward pod/nodejs-pod 3000:3000  
Forwarding from 127.0.0.1:3000 -> 3000  
Forwarding from [::1]:3000 -> 3000
```

Step 7: Open browser

<http://localhost:3000>

